

```
dvdrental=# CREATE SCHEMA data_mart;
CREATE SCHEMA
dvdrental=# CREATE SCHEMA operations;
CREATE SCHEMA
dvdrental=# CREATE SCHEMA analytics;
CREATE SCHEMA
dvdrental=# CREATE SCHEMA finance;
CREATE SCHEMA
dvdrental=# CREATE SCHEMA management;
CREATE SCHEMA
dvdrental=# \dn
```

List of schemas

Name	Owner
analytics	postgres
data_mart	postgres
finance	postgres
management	postgres
operations	postgres
public	pg_database_owner

(6 rows)

```
dvdrental=# |
```

```
PS C:\Users\Administrador\Documents\GitHub\dba_postgresql> Get-Content "C:\Users\Administrador\Documents\GitHub\dba_postgresql\sql\migracao_data_mart.sql" | docker exec -i postgresql psql -U postgres -d dvdren
tal
DO
PS C:\Users\Administrador\Documents\GitHub\dba_postgresql> docker exec -it postgresql bash
root@01ff8f25a820:/# psql -U postgres -d dvdrental
psql (17.5 (Debian 17.5-1.pgdg120+1))
Type "help" for help.

dvdrental=# \dC
dvdrental=# \dn+

      Name      | Owner      | List of schemas | Access privileges | Description
-----
analytics       | postgres   |                  |                   |
data_mart       | postgres   |                  |                   |
finance         | postgres   |                  |                   |
management      | postgres   |                  |                   |
operations      | postgres   |                  |                   |
public          | pg_database_owner | pg_database_owner=UC/pg_database_owner+ | =U/pg_database_owner | standard public schema
(6 rows)

dvdrental=# \dt
Did not find any relations.
dvdrental=# SHOW search_path;
search_path
-----
"$user", public
(1 row)

dvdrental=# SET search_path TO data_mart;
SET
dvdrental=# SHOW search_path;
search_path
-----
data_mart
(1 row)

dvdrental=# \dt
      Schema      | List of relations |
-----
Schema | Name | Type | Owner
-----
data_mart | actor | table | postgres
data_mart | address | table | postgres
data_mart | category | table | postgres
data_mart | city | table | postgres
data_mart | country | table | postgres
data_mart | customer | table | postgres
data_mart | film | table | postgres
data_mart | film_actor | table | postgres
data_mart | film_category | table | postgres
```

```

dvdrental=# CREATE ROLE dm_read_access NOLOGIN;
CREATE ROLE
dvdrental=# CREATE ROLE dm_sequence_usage NOLOGIN;
CREATE ROLE
dvdrental=# CREATE ROLE ops_rental_write_access NOLOGIN;
CREATE ROLE
dvdrental=# CREATE ROLE ops_customer_write_access NOLOGIN;
CREATE ROLE
dvdrental=# CREATE ROLE fin_read_access NOLOGIN;
CREATE ROLE
dvdrental=# CREATE anl_read_access NOLOGIN;
ERROR:  syntax error at or near "anl_read_access"
LINE 1: CREATE anl_read_access NOLOGIN;
              ^

dvdrental=# CREATE ROLE anl_read_access NOLOGIN;
CREATE ROLE
dvdrental=# CREATE ROLE mgmt_read_access NOLOGIN;
CREATE ROLE
dvdrental=# CREATE ROLE job_cashier NOLOGIN;
CREATE ROLE
dvdrental=# CREATE ROLE job_store_manager NOLOGIN;
CREATE ROLE
dvdrental=# CREATE ROLE job_data_analyst NOLOGIN;
CREATE ROLE
dvdrental=# CREATE ROLE job_finance_director NOLOGIN;
CREATE ROLE
dvdrental=# CREATE ROLE job_owner NOLOGIN;
CREATE ROLE
dvdrental=# CREATE ROLE service_accounts NOLOGIN;
CREATE ROLE
dvdrental=# CREATE ROLE pedro_caixa WITH LOGIN PASSWORD 'senha_forte_para_pedro';
CREATE ROLE
dvdrental=# CREATE ROLE maria_gerente WITH LOGIN PASSWORD 'senha_forte_para_maria';
CREATE ROLE
dvdrental=# CREATE ROLE ana_analista WITH LOGIN PASSWORD 'senha_forte_para_ana';
CREATE ROLE
dvdrental=# CREATE ROLE carlos_diretor WITH LOGIN PASSWORD 'senha_forte_para_carlos';
CREATE ROLE
dvdrental=# CREATE ROLE helena_dona WITH LOGIN PASSWORD 'senha_forte_para_helena';
CREATE ROLE
dvdrental=# CREATE ROLE app_service WITH LOGIN PASSWORD 'senha_muito_longa_e_complexa_para_app';
CREATE ROLE
dvdrental=# |

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dvdrental=# CREATE ROLE app_service WITH LOGIN PASSWORD 'senha_muito_longa_e_complexa_para_app';
CREATE ROLE
dvdrental=# GRANT dm_read_access TO job_cashier, job_store_manager, job_data_analyst, job_finance_director, job_owner, service_accounts;
GRANT ROLE
dvdrental=# GRANT dm_sequence_usage TO job_cashier, job_store_manager, service_accounts;
GRANT ROLE
dvdrental=# GRANT ops_rental_write_access TO job_cashier, job_store_manager;
GRANT ROLE
dvdrental=# GRANT ops_customer_write_access TO job_store_manager;
GRANT ROLE
dvdrental=# GRANT fin_read_access TO job_finance_director, job_owner;
GRANT ROLE
dvdrental=# GRANT anl_read_access TO job_data_analyst, job_store_manager, job_owner;
GRANT ROLE
dvdrental=# GRANT mgmt_read_access TO job_store_manager, job_owner;
GRANT ROLE
dvdrental=# GRANT job_cashier TO pedro_caixa;
GRANT ROLE
dvdrental=# GRANT job_store_manager TO maria_gerente;
GRANT ROLE
dvdrental=# GRANT job_data_analyst TO ana_analista;
GRANT ROLE
dvdrental=# GRANT job_finance_director TO carlos_diretor;
GRANT ROLE
dvdrental=# GRANT job_owner TO helena_dona;
GRANT ROLE
dvdrental=# GRANT service_accounts TO app_service;
GRANT ROLE
dvdrental=# |

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Windows PowerShell
X + v

dvdrental=# SHOW search_path;
search_path
-----
data_mart
(1 row)

dvdrental=# SET search_path TO operations;
SET
dvdrental=# SHOW search_path;
search_path
-----
operations
(1 row)

dvdrental=# CREATE VIEW operations.new_rental AS
dvdrental=# SELECT film_id, customer_id, store_id FROM data_mart.rental;
ERROR: column "film_id" does not exist
LINE 2: SELECT film_id, customer_id, store_id FROM data_mart.rental;
              ^
dvdrental=# SELECT * FROM data_mart.rental;
dvdrental=# SELECT * FROM data_mart.film;
dvdrental=# SELECT * FROM data_mart.store;
store_id | manager_staff_id | address_id | last_update
-----
1 | 1 | 1 | 2006-02-15 09:57:12
2 | 2 | 2 | 2006-02-15 09:57:12
(2 rows)

dvdrental=# CREATE VIEW operations.new_rental AS
dvdrental=# SELECT * FROM data_mart.film;
CREATE VIEW
dvdrental=# \dv
      List of relations
 Schema | Name | Type | Owner
-----
operations | new_rental | view | postgres
(1 row)

dvdrental=# CREATE VIEW analytics.film_performance_summary AS
dvdrental=# SELECT * FROM data_mart.sales_by_film_category;
CREATE VIEW
dvdrental=# \dv
      List of relations
 Schema | Name | Type | Owner
-----
operations | new_rental | view | postgres
(1 row)

dvdrental=# SET search_path TO analytics;
SET
```

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Windows PowerShell
X + v

dvdrental=# CREATE VIEW operations.new_rental AS
dvdrental=# SELECT * FROM data_mart.film;
CREATE VIEW
dvdrental=# \dv
      List of relations
 Schema | Name | Type | Owner
-----
operations | new_rental | view | postgres
(1 row)

dvdrental=# CREATE VIEW analytics.film_performance_summary AS
dvdrental=# SELECT * FROM data_mart.sales_by_film_category;
CREATE VIEW
dvdrental=# \dv
      List of relations
 Schema | Name | Type | Owner
-----
operations | new_rental | view | postgres
(1 row)

dvdrental=# SET search_path TO analytics;
SET
dvdrental=# SHOW search_path;
search_path
-----
analytics
(1 row)

dvdrental=# \dv
      List of relations
 Schema | Name | Type | Owner
-----
analytics | film_performance_summary | view | postgres
(1 row)

dvdrental=# CREATE VIEW finance.daily_revenue_summary AS
dvdrental=# SELECT payment_date::date AS day, sum(amount) AS total_revenue
dvdrental=# FROM data_mart.payment
dvdrental=# GROUP BY day
dvdrental=# ORDER BY day DESC;
CREATE VIEW
dvdrental=# SET search_path TO finance;
SET
dvdrental=# \dv
      List of relations
 Schema | Name | Type | Owner
-----
finance | daily_revenue_summary | view | postgres
(1 row)

dvdrental=# |
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Windows PowerShell
x + v

dvdrental=# \dv
Schema | List of relations | Type | Owner
-----|-----|-----|-----
analytics | film_performance_summary | view | postgres
(1 row)

dvdrental=# CREATE VIEW finance.daily_revenue_summary AS
dvdrental=# SELECT payment_date:date AS day, sum(amount) AS total_revenue
dvdrental=# FROM data_mart.payment
dvdrental=# GROUP BY day
dvdrental=# ORDER BY day DESC;
CREATE VIEW
dvdrental=# SET search_path TO finance;
SET
dvdrental=# \dv
Schema | List of relations | Type | Owner
-----|-----|-----|-----
finance | daily_revenue_summary | view | postgres
(1 row)

dvdrental=# CREATE VIEW management.store_performance AS
dvdrental=# SELECT * FROM data_mart.sales_by_store;
CREATE VIEW
dvdrental=# SET search_path TO management;
SET
dvdrental=# \dv
Schema | List of relations | Type | Owner
-----|-----|-----|-----
management | store_performance | view | postgres
(1 row)

dvdrental=# GRANT USAGE ON SCHEMA data_mart, operations, analytics, finance, management
dvdrental=# TO dm_read_access;
GRANT
dvdrental=# GRANT SELECT ON ALL TABLES IN SCHEMA data_mart TO dm_read_access;
GRANT
dvdrental=# GRANT USAGE ON ALL SEQUENCES IN SCHEMA data_mart TO dm_sequence_usage;
GRANT
dvdrental=# GRANT INSERT, UPDATE ON operations.new_rental TO ops_rental_write_access;
GRANT
dvdrental=# GRANT SELECT ON ALL TABLES IN SCHEMA analytics TO anl_read_access;
GRANT
dvdrental=# GRANT SELECT ON ALL TABLES IN SCHEMA finance TO fin_read_access;
GRANT
dvdrental=# GRANT SELECT ON ALL TABLES IN SCHEMA management TO mgmt_read_access;
GRANT
dvdrental=# |
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```
Windows PowerShell
x + v

dvdrental=# \q
root@091ff8f25a820:/# psql -U pedro_caixa -d dvdrental
psql (17.5 (Debian 17.5-1.pgdg120+1))
Type "help" for help.

dvdrental=> \du+
Role name | List of roles | Attributes | Description
-----|-----|-----|-----
ana_analista | | | Cannot login
anl_read_access | | | Cannot login
app_service | | | Cannot login
carlos_diretor | | | Cannot login
dm_read_access | | | Cannot login
dm_sequence_usage | | | Cannot login
fin_read_access | | | Cannot login
helena_dona | | | Cannot login
job_cashier | | | Cannot login
job_data_analyst | | | Cannot login
job_finance_director | | | Cannot login
job_owner | | | Cannot login
job_store_manager | | | Cannot login
maria_gerente | | | Cannot login
mgmt_read_access | | | Cannot login
ops_customer_write_access | | | Cannot login
ops_rental_write_access | | | Cannot login
pedro_caixa | | | Superuser, Create role, Create DB, Replication, Bypass RLS
postgres | | | Cannot login
service_accounts | | | Cannot login

dvdrental=> SELECT title FROM data_mart.film WHERE film_id = 10;
title
-----
Aladdin Calendar
(1 row)

dvdrental=> SELECT * FROM operations.new_rental LIMIT 10;
ERROR: permission denied for view new_rental
dvdrental=> SHOW search_path;
search_path
-----
"$user", public
(1 row)

dvdrental=> SELECT current_user;
current_user
-----
pedro_caixa
(1 row)

dvdrental=> |
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Windows PowerShell
dvdrental=> SELECT current_user;
current_user
-----
pedro_caixa
(1 row)

dvdrental=> INSERT INTO operations.new_rental (rental_date, inventory_id, customer_id, return_date, staff_id, last_update)
VALUES (NOW(), 2346, 142, NULL, 1, NOW());
ERROR: column "rental_date" of relation "new_rental" does not exist
LINE 1: INSERT INTO operations.new_rental (rental_date, inventory_id...
                                         ^

dvdrental=> DROP VIEW operations.new_rental;
ERROR: must be owner of view new_rental
dvdrental=> exit
root@01ff8f25a820:/# psql -U postgres -d dvdrental
psql (17.5 (Debian 17.5-1.pgdg120+1))
Type "help" for help.

dvdrental=# DROP VIEW operations.new_rental;
DROP VIEW
dvdrental=# CREATE VIEW operations.new_rental AS
dvdrental=# SELECT * FROM data_mart.rental;
CREATE VIEW
dvdrental=# EXIT
root@01ff8f25a820:/# psql -U pedro_caixa -d dvdrental
psql (17.5 (Debian 17.5-1.pgdg120+1))
Type "help" for help.

dvdrental=> GRANT INSERT, UPDATE ON operations.new_rental TOP ops_rental_write_access;
ERROR: syntax error at or near "TOP"
LINE 1: GRANT INSERT, UPDATE ON operations.new_rental TOP ops_rental...
                                         ^

dvdrental=> GRANT INSERT, UPDATE ON operations.new_rental TO ops_rental_write_access;
ERROR: permission denied for table new_rental
dvdrental=> exit
root@01ff8f25a820:/# psql -U postgres -d dvdrental
psql (17.5 (Debian 17.5-1.pgdg120+1))
Type "help" for help.

dvdrental=# GRANT INSERT, UPDATE ON operations.new_rental TOP ops_rental_write_access;
ERROR: syntax error at or near "TOP"
LINE 1: GRANT INSERT, UPDATE ON operations.new_rental TOP ops_rental...
                                         ^

dvdrental=# GRANT INSERT, UPDATE ON operations.new_rental TO ops_rental_write_access;
GRANT
dvdrental=# exit
root@01ff8f25a820:/# psql -U pedro_caixa -d dvdrental
psql (17.5 (Debian 17.5-1.pgdg120+1))
Type "help" for help.

dvdrental=> |

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Windows PowerShell
dvdrental=> INSERT INTO operations.new_rental (rental_date, inventory_id, customer_id, return_date, staff_id, last_update)
VALUES (NOW(), 2346, 142, NULL, 1, NOW());
INSERT 0 1
dvdrental=> SELECT * FROM finance.daily_revenue_summary;
ERROR: permission denied for view daily_revenue_summary
dvdrental=> exit
root@01ff8f25a820:/# psql -U ana_analista -d dvdrental
psql (17.5 (Debian 17.5-1.pgdg120+1))
Type "help" for help.

dvdrental=> SELECT * FROM analytics.film_performance_summary;
 category | total_sales
-----+-----
 Sports   | 4892.19
 Sci-Fi   | 4336.01
 Animation | 4245.31
 Drama    | 4118.46
 Comedy   | 4002.48
 New      | 3966.38
 Action   | 3951.84
 Foreign  | 3934.47
 Games    | 3922.18
 Family   | 3830.15
 Documentary | 3749.65
 Horror   | 3401.27
 Classics | 3353.38
 Children | 3309.39
 Travel   | 3227.36
 Music    | 3071.52
(16 rows)

dvdrental=> INSERT INTO operations.new_rental (rental_date, inventory_id, customer_id, return_date, staff_id, last_update)
VALUES (NOW(), 2346, 130, NULL, 2, NOW());
ERROR: permission denied for view new_rental
dvdrental=> EXIT
root@01ff8f25a820:/# psql -U helena_dona -d dvdrental
psql (17.5 (Debian 17.5-1.pgdg120+1))
Type "help" for help.

dvdrental=> \du+ dona_helena
List of roles
Role name | Attributes | Description
-----+-----+-----
dvdrental=> \du+ helena_dona
List of roles
Role name | Attributes | Description
-----+-----+-----
helena_dona | |
dvdrental=>

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```
Windows PowerShell
helena_dona |
dvdrental=> \/+
List of databases
Name | Owner | Encoding | Locale Provider | Collate | Ctype | Locale | ICU Rules | Access privileges | Size | Tablespace | Description
-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----
dvdrental | postgres | UTF8 | libc | en_US.utf8 | en_US.utf8 | | | | 15 MB | pg_default | default administrative connection database
postgres | postgres | UTF8 | libc | en_US.utf8 | en_US.utf8 | | | | 7475 kB | pg_default | unmodifiable empty database
template0 | postgres | UTF8 | libc | en_US.utf8 | en_US.utf8 | | | | postgres=CtC/postgres + 7321 kB | pg_default |
template1 | postgres | UTF8 | libc | en_US.utf8 | en_US.utf8 | | | | postgres=CtC/postgres + 7547 kB | pg_default | default template for new databases
(4 rows)

dvdrental=> \dp
Access privileges
Schema | Name | Type | Access privileges | Column privileges | Policies
-----|-----|-----|-----|-----|-----
(0 rows)

dvdrental=> SELECT * FROM daily_revenue_summary;
ERROR: relation "daily_revenue_summary" does not exist
LINE 1: SELECT * FROM daily_revenue_summary;
^
dvdrental=> SELECT * FROM finance.daily_revenue_summary;
day | total_revenue
-----|-----
2007-05-14 | 514.18
2007-04-30 | 5723.89
2007-04-29 | 2717.60
2007-04-28 | 2622.73
2007-04-27 | 2673.57
2007-04-26 | 347.21
2007-04-12 | 1930.48
2007-04-11 | 1940.32
2007-04-10 | 1973.18
2007-04-09 | 2067.86
2007-04-08 | 2227.84
2007-04-07 | 1984.28
2007-04-06 | 2077.14
2007-04-05 | 273.36
2007-03-23 | 2342.43
2007-03-22 | 2586.79
2007-03-21 | 2868.27
2007-03-20 | 2669.89
2007-03-19 | 2617.69
2007-03-18 | 2701.76
2007-03-17 | 2442.16
2007-03-16 | 299.28
2007-03-02 | 2550.05
```

```
Windows PowerShell
PS C:\Users\Administrador\Documents\GitHub\dba_postgresql> Get-Content "C:\Users\Administrador\Documents\GitHub\dba_postgresql\sql\ownership_query.sql" | docker exec -i postgresql psql -U postgres -d dvdrental
Schema | Name | Type | Owner
-----|-----|-----|-----
(0 rows)

PS C:\Users\Administrador\Documents\GitHub\dba_postgresql> Get-Content "C:\Users\Administrador\Documents\GitHub\dba_postgresql\sql\ownership_query.sql" | docker exec -i postgresql psql -U postgres -d dvdrental
Schema | Name | Type | Owner
-----|-----|-----|-----
analytics | film_performance_summary | view | postgres
data_mart | actor | table | postgres
data_mart | actor_actor_id_seq | sequence | postgres
data_mart | actor_info | view | postgres
data_mart | address | table | postgres
data_mart | address_address_id_seq | sequence | postgres
data_mart | category | table | postgres
data_mart | category_category_id_seq | sequence | postgres
data_mart | city | table | postgres
data_mart | city_city_id_seq | sequence | postgres
data_mart | country | table | postgres
data_mart | country_country_id_seq | sequence | postgres
data_mart | customer | table | postgres
data_mart | customer_customer_id_seq | sequence | postgres
data_mart | customer_list | view | postgres
data_mart | film | table | postgres
data_mart | film_actor | table | postgres
data_mart | film_category | table | postgres
data_mart | film_film_id_seq | sequence | postgres
data_mart | film_list | view | postgres
data_mart | inventory | table | postgres
data_mart | inventory_inventory_id_seq | sequence | postgres
data_mart | language | table | postgres
data_mart | language_language_id_seq | sequence | postgres
data_mart | nicer_but_slower_film_list | view | postgres
data_mart | payment | table | postgres
data_mart | payment_payment_id_seq | sequence | postgres
data_mart | rental | table | postgres
data_mart | rental_rental_id_seq | sequence | postgres
data_mart | sales_by_film_category | view | postgres
data_mart | sales_by_store | view | postgres
data_mart | staff | table | postgres
data_mart | staff_list | view | postgres
data_mart | staff_staff_id_seq | sequence | postgres
data_mart | store | table | postgres
data_mart | store_store_id_seq | sequence | postgres
finance | daily_revenue_summary | view | postgres
management | store_performance | view | postgres
operations | new_rental | view | postgres
(39 rows)
```